

Conformal Prediction is Early Stopping

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Abstract

Split conformal prediction ends where its calibration begins: fit a model, pool its held-out residuals, quote the pooled quantiles, stop. Of 686 arXiv papers with conformal prediction in the title, 599 stop at that empirical map, and the one candidate for continuing past it, on reading, also stops. We show the stop is the expensive part. An identity pins the irreducible excess log score of a frozen scheme to its score and stratification; on 17,364 economic series, stopping costs 0.40 to 0.48 nats after a null-control adjustment, composing past the map recovers a third to a half, and empirical coverage detects none of it.

1 Introduction

Conformal prediction attaches a guarantee to any point forecaster by a recipe. Fit a model however you like. On n held-out points, compute a scalar error for each, for instance the absolute residual. Sort the errors, take the $\lceil (n+1)0.9 \rceil$ -th smallest, and quote the model's prediction plus or minus that value. If the held-out data and the future point are exchangeable, meaning their joint law is unchanged by shuffling, the quoted interval contains the truth at least 90% of the time, at any sample size, with no assumption on the model or the data distribution.

The recipe ends with the percentile. Nothing is ever fitted to the output of the pooled-error step: the interval it produces is the answer. In machine learning, early stopping is a deliberate act, a regularizer chosen because validation loss says further fitting would hurt. The conformal stop is not chosen against anything. It is where the recipe happens to end, and as far as we can determine no paper argues for it.

This paper treats the stop as an empirical question. The unit throughout is the nat, the natural unit of the logarithmic score: a forecaster whose average log density trails another's by 0.15 nats assigns the realized outcomes $e^{0.15} \approx 1.16$ times less probability per observation. Beside it we report CRPS, the continuous ranked probability score, which integrates the squared difference between the forecast's cumulative distribution and the outcome's step function; it is in the units of the data, lower is better, and it is the score most conformal papers report.

The paper has three components.

1. *The count* (Section 2). In a frame of 686 arXiv papers, 599 stop at the empirical map. Papers that attack the pooled quantile replace it or reweight it; none keeps it and continues. The single flagged candidate for continuing, a PNAS paper, conditions before the map and stops after it.

2. *The floor* (Section 3). For a class of frozen conformal schemes, the excess log score over the conditional oracle decomposes exactly into a conditional mutual information fixed by the score and stratification, plus an estimation term. Calibration choices move only the estimation term. The identity says what stopping protects and what it abandons.
3. *The price* (Section 4). Inside a transform grammar where the empirical map is one invertible step among peers, we hold one forecaster and one empirical object fixed and move only the object’s position: last, as the convention has it, or interior, with modelling continuing on its output. On 17,364 economic series, stopping costs 0.40 to 0.48 nats after a null-control adjustment; the interior placement recovers a third to a half of that; and the stopped arm’s empirical coverage sits nearer the nominal target than the unconstrained base’s, so coverage monitoring is blind to the entire loss.

Section 5 states the recommendation: keep the map, keep the certificate where a certificate is wanted, and keep modelling past it.

What is new and what is borrowed. The count and the price are this paper’s, and the floor identity is proved in Appendix A with its scope stated carefully. The published theory around the pooled shape is context, not a contribution here: exact distribution-free conditional coverage is impossible in nonatomic settings (Lei and Wasserman, 2014; Barber et al., 2021); weighting calibration points unequally costs coverage in proportion to the total-variation distance between their residual laws (Barber et al., 2023); and a guarantee held at a fixed point rather than on average inflates the threshold for classical testing-power reasons (Li, 1989; Low, 1997; Cai and Low, 2004). Together these give the pooled shape a defence. Nothing in them concerns the stop, which is the point of this paper. The base forecaster is the subject of Cotton (2026).

2 The count

The claim that everyone stops concerns published work, so it gets a protocol: a defined frame, documented keyword rules, and full-text reading of every paper whose label could change the conclusion. The frame is every arXiv paper with “conformal prediction” in the title at harvest date, $n = 686$. Classifier, frame, and per-paper verdicts are in the repository.

The object classified is the *empirical map*: the step turning the multiset of held-out errors into a distribution, concretely the empirical cumulative distribution of the calibration scores. A paper *stops* if the output of that map, its quantiles, its p -values, or its intervals, is the answer. The alternatives found in the frame: *replace* the pooled quantile with a conditional estimate (SPCI fits a quantile regression to recent residuals (Xu and Xie, 2023)); *tune* the level online while keeping the map (adaptive conformal inference (Gibbs and Candès, 2021)); *consume* the output in a downstream decision such as screening pseudo-labels; or *compose*: keep the map and fit a model to its output, the move whose absence is the finding.

Of 686 papers, 599 stop, 16 replace, 28 tune, 38 consume, and 5 propose no method. One compose candidate was flagged by the screen: distributional conformal prediction (Chernozhukov et al., 2021), which uses an estimated conditional distribution as its score. Read in full, it conditions *before* the map, and the answer is the empirical quantile of the resulting ranks: a richer score, then a stop. The refutation criterion for the count is stated and falsifiable: a paper that retains the empirical rank transform, applies it to residuals, and fits a downstream model on

the transformed stream. Reweighting the empirical law does not count, conditioning upstream does not count, adapting the level does not count.

Outside the conformal banner the composing move is routine. The normal quantile transform in hydrological forecasting (Bogner et al., 2012), copula time-series models (Chen and Fan, 2006), and autoregressive transformation models (Rügamer et al., 2023) all apply an empirical marginal map and keep modelling in the transformed coordinates. The absence is a habit of one community, not a difficulty of the idea.

A second classification, reported in the repository, checks how papers treat residual pooling: 94.5% never let the weights on calibration residuals depend on the test point at any granularity, and each of the nine smooth-weighting papers that claims exact validity was read in full and found to claim it through a legitimate escape (marginal recalibration, a stated coverage gap, a known covariate-shift tilt, or added smoothness assumptions). The field polices the pooling constraint carefully. The stop it does not police at all.

3 The floor

The identity below says what the calibration step can and cannot influence. It needs two definitions and has a deliberately narrowed scope.

A *nonconformity score* is a map $A : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$ assigning a scalar to each candidate outcome y at input x . A *stratification* $\kappa : \mathcal{X} \rightarrow \mathcal{K}$ assigns each input to a category. Both are fit on a training fold and frozen, since exchangeability of calibration and test scores is what the coverage argument uses. The schemes covered by the proposition are those whose score is, for each x , a strictly increasing absolutely continuous bijection in y : signed-residual and normalized signed-residual schemes, and predictive systems built on invertible score transformations. Folded scores such as the absolute residual $|y - \hat{\mu}(x)|$ and quantile-regression scores are not bijections in y and need a separate treatment; the empirical arms in Section 4 are of the covered type.

Such a scheme emits the pooled empirical law of calibration scores in the test point’s stratum, pushed back through $A(x, \cdot)^{-1}$. For log scoring the emitted law must have a density, so let $\hat{G}_\kappa$ be an absolutely continuous estimate of the pooled score law (a kernel-smoothed or interpolated empirical law; the raw empirical measure is atomic and scores $-\infty$). Write $W = A(X, Y)$, let G_κ be the population pooled law, and let the benchmark be the conditional oracle, whose expected negative log score is the conditional entropy $H(Y | X)$. Throughout, $L(\hat{F}) = \mathbb{E}[-\log \hat{f}_X(Y)]$ is the expected log loss of the conformal predictive, so an excess is positive and smaller is better.

Proposition 1 (The floor, exactly). *Fix the training fold and the calibration sample, so $(A, \kappa, \hat{G}_\kappa)$ are deterministic, let (X, Y) be a test pair independent of both, and assume the laws below admit densities with finite divergences. Then the excess log loss of the conformal predictor decomposes exactly as*

$$L(\hat{F}) - H(Y | X) = I(W; X | \kappa) + \mathbb{E}_\kappa \text{KL}(G_\kappa \| \hat{G}_\kappa),$$

where $I(W; X | \kappa)$ is the conditional mutual information between score and input given the stratum. (Unconditionally, take the outer expectation over the calibration sample; the first term is unchanged.) The second term is nonnegative and is the only part any calibration choice, estimator, quantile convention, randomization, or level, can affect. At a fixed calibration sample it vanishes if and only if $\hat{G}_\kappa = G_\kappa$ almost everywhere; KL-consistent estimation of the pooled law with growing calibration mass per stratum drives it to zero asymptotically, which is an assumption on the estimator, not a consequence of the setup.

The proof is a change of variables in which the Jacobian of A cancels between truth and forecast, then a split at the population pooled law (Appendix A). In words: the input reaches the predictive only through the frozen reshaping and the frozen stratum choice, so whatever information about the score the input still carries within a stratum is lost before calibration begins, and no amount of calibration data buys it back. With a location score and one stratum the floor is $I(R; X)$, the mutual information between the model’s error and its input.

Remark 2. The floor is zero exactly when the frozen pair renders the score independent of the input within strata: $I(W; X | \kappa) = 0$ iff $W \perp X | \kappa$. When X carries the full information set available downstream, that is the statement that the modelling was finished before the conformal step. Conformal calibration is especially valuable when one is unwilling to assume that; but its guarantee does not diagnose whether the floor is zero, and one may deploy it on perfectly finished residuals purely for finite-sample validity. What the identity rules out is the middle ground: whenever the floor is positive, no calibration choice recovers any of it.

The floor explains what stopping protects: the certificate, which concerns the marginal rank of the test score and is genuinely exact at its position. It also says what stopping abandons: $I(W; X | \kappa)$, the conditional information the frozen score discarded. A downstream model consuming the map’s output, together with the input, is one means of recovering part of that quantity, not an additional source of loss beyond it. The published results cited in Section 1 defend the pooling constraint inside the certificate. No result defends ending the pipeline. Whether ending it is expensive is a measurement question, taken up next.

4 The price

4.1 A laboratory where position is a variable

Inside standard conformal practice the stop is invisible: the pipeline ends at the map by construction, so there is nothing to vary. To measure the stop one needs a setting where the conformal step is an ordinary component, and that requires seeing it as what Section 3 showed it to be: a univariate transform. The raw empirical map is a step function, not invertible and without a density; repaired by strictly monotone interpolation through its knots, with kernel-smoothed mass so it can be log-scored, it becomes an invertible state-updated map of the observation coordinate, and such a map can sit anywhere in a chain.

The laboratory is a transform grammar (Cotton, 2026). A forecaster is a *chain* of state-conditional invertible transforms, each removing one kind of structure and passing its residual on: a location tracker subtracts a running mean, a scale transform divides by a running deviation, a difference removes trend. A *terminal leaf* models the law of whatever residual remains, and the predictive in the original coordinate is the leaf’s law pushed back through the inverse chain, change of variables handled exactly. The base forecaster, `laplace`, runs roughly sixty candidate chains in parallel, combines their predictive densities with likelihood weights updated every observation, splices generalized Pareto tails onto the extremes, and scores its own probability integral transforms online. It ranks first in the public benchmark this study extends, and its own terminal leaf is tuned to CRPS.

In this laboratory the conformal habit is one configuration among two:

stop. Take the finished predictive, keep its centre, replace its shape with the pooled empirical law of the model’s own recent errors, kernel-smoothed. Quote that. Nothing acts

downstream. This arm is a frozen conformal scheme of the covered type, so Proposition 1 applies with floor $I(R; X)$.

compose-after. Install the identical empirical object as the terminal leaf of *every* candidate chain. Writing R_t^c for chain c ’s residual and $\hat{G}_t^c$ for the pooled empirical law of that chain’s own residual stream, the arm’s predictive is

$$\hat{F}_t = \sum_c w_t^c (T_t^c)_\#^{-1} \hat{G}_t^c,$$

where $(T_t^c)_\#^{-1}$ pushes the map’s law back through chain c ’s inverse transform and the weights w_t^c are likelihood weights fitted online to the realized densities the maps assign, that is, a mixture model fitted on the maps’ output stream. Every predictive path passes through an empirical map: the leaf override replaces the terminal leaf of every candidate chain in the implementation, so the pool weights transform chains against each other, never the map against a parametric alternative, and no weight can route around it. The one exception is the generalized Pareto tail splice, which replaces the density beyond a frozen tail region; it acts on the extremes of the composed predictive and is shared with the base arm. This arm carries no coverage certificate. It is the move the floor identity permits, a change of representation, executed after the map rather than before it.

Both positions run flat and recency-weighted (exponential weights, half-life 150 observations), so staleness explains no comparison.

4.2 Design

The corpus is FRED economic change series (differences of rates, production, employment and similar), non-price, with two tiers. The *breadth* tier uses 256 observations of history per forecast origin and every qualifying series: 17,364. The *deep* tier uses 1,024 observations and series with at least 1,100, so the 400-residual window is a strict subset of the history it could have used: about 6,400 series. Twelve rolling origins per series, horizons 1, 6 and 12. Because origins within a series overlap, all intervals are bootstrap over series, not rows.

A control corpus of synthetic independent draws (Gaussian, Student- t_3 , Laplace) runs under the identical protocol at matched context. On the control there is no state to condition on, so what a constrained arm loses there is finite-sample overhead: the cost of estimating a 41-node empirical law rather than a parametric scale. That overhead is subtracted from every headline number below, and we call the result *null-control-adjusted* rather than net of estimation, for two reasons the adjustment cannot fix. The estimation term $\text{KL}(G \parallel \hat{G})$ depends on G , so its size on three simple iid laws is a gauge, not an unbiased estimate of its size on economic residuals; and because the base forecaster adapts, its residuals are not exactly state-independent even on iid data. Raw deficits, without the adjustment, are 0.51, 0.61 and 0.59 nats at horizons 1, 6 and 12 on the breadth tier (paired, 208,368 rows per horizon): the adjustment removes about a fifth of the raw gap, and both versions are reported so no reader has to trust the gauge. A sharper control, permuting each real series so the marginal residual law is preserved while temporal state dependence is destroyed; where the two gauges disagree, the permutation control governs.

4.3 The decomposition, null-control-adjusted

Breadth tier, 17,364 series, null-control-adjusted, log score deficit against the unconstrained base:

	$h=1$	$h=6$	$h=12$
stop, total cost	0.397	0.475	0.462
of which pooled shape	0.267	0.315	0.275
of which position	0.131	0.160	0.187

Raw per-arm deficits carry series-bootstrap 95% intervals of about ± 0.02 throughout. Composing recovers 33%, 34% and 40% of the cost at the three horizons. The deep tier gives raw stop costs of 0.34 to 0.40 nats with 40 to 55 percent recovered by the interior placement.

The deep-tier fraction is not yet a pure position effect, and we flag the confound rather than interpret past it. Between tiers, three things change at once: 256 against 1,024 observations of context, the series panels differ, and the 400-residual window truncates a 1,024-observation history while holding all of a 256-observation one. Nor is a 400-observation window inherent to stopping; a stopped map may retain every residual it has ever seen. The clean separation is a same-series, same-context ablation in which only one bit changes at a time: stop arms at windows 250, 400, 750 and all history, and composed arms matched window for window, on the ctx-1024 panel. Until that ablation reports, the hypothesis that *the price of stopping grows with how much the pooled window discards* is exactly that, a hypothesis the current tiers are consistent with but do not identify.

The arms are close on the median series; the mean cost is carried by a minority of series where stopping hurts badly. Stopping behaves like the removal of insurance.

Adaptivity does not substitute for position. Recency-weighting the stop arm’s window recovers 0.05 to 0.08 nats on the breadth tier, against 0.14 to 0.27 for moving the map’s position, raw in both cases; and level adaptation is dispatched by Proposition 1 directly, since it acts on the pooled scores alone.

On CRPS the tiers disagree: on the deep tier the stopped arm degrades CRPS and the composed arm improves it, while on the breadth tier both positions sit within about 1% of the base. The disagreement is reported, not resolved, and CRPS appears in no summary of this paper.

4.4 Empirical coverage is blind to the loss

Coverage is the axis conformal practice monitors, so we measure it directly, with two scope notes. These are economic time series and we assert no exchangeability, so what follows is empirical coverage, not the finite-sample certificate of Section 1; and the intervals are the 0.1 and 0.9 quantiles of the kernel-smoothed predictive, which need not coincide with the raw conformal order-statistic endpoints. What the table tests is therefore the monitoring practice: whether watching interval coverage would reveal the log score loss. Breadth tier, paired rows, target 0.800:

arm	coverage	distance from target
unconstrained	0.870	0.070
stop	0.784	0.016
stop, weighted	0.799	0.001
compose-after	0.797	0.003

We expected the stopped arm to show good overall coverage, and it does, along with something the log score column would never suggest: every conformal arm sits nearer nominal than

the unconstrained base, which overcovers at 0.870. On this axis the conformal arms win. On the log score axis the stopped arm loses 0.40 to 0.48 nats. Both statements are true at once, which is the point: coverage tracks the rank of the outcome in the predictive; the floor tracks conditional information the frozen score discarded; the two move independently, and a practice that monitors only coverage will never see the loss.

5 Discussion

The title is a description. The conformal recipe stops at its empirical map the way training stops at an epoch: except that in learning the stop is chosen against validation evidence, and here it is chosen against nothing. Measured, the stop costs 0.40 to 0.48 nats after a null-control adjustment on more than seventeen thousand series, and composing an ordinary model past the map recovers a third to a half of that at no cost to empirical coverage. The floor identity locates the part of the loss that is fixed before calibration begins, and Remark 2 states the corollary that matters in practice: whenever the floor is positive, and the method is most attractive exactly when one cannot rule that out, no calibration choice recovers any of it.

The recommendation is not a better calibrator, since every calibration choice moves only the estimation term. It is to treat the empirical map as a coordinate change with a certificate attached at its position, and to keep modelling on its output. The composed arm is one constructive instance, and in the sense of Remark 2 it is conformal prediction with the modelling finished; its scored calibration stream then measures, rather than assumes, whether the modelling ever was.

Limitations. The count is a keyword screen with full-text verification of every label that could change the conclusion; the 599 stop default rests on the screen plus a stated refutation criterion. The prices are measured on one base forecaster and one economic corpus plus a synthetic control; the deep tier and the matched-window ablation of Section 4.3 are incomplete. The null-control adjustment gauges finite-sample overhead on simple iid laws and is not an unbiased estimate of the estimation term on economic residuals. Scoring floors the log density at -25 per observation, which is not a proper score where it binds, and it binds unequally: on 1.78% of stopped-arm rows against 0.62% of base rows (breadth tier, paired). The inequality favours the stopped arm, and tightening the floor stepwise from -25 to -10 shrinks the raw stop deficit monotonically from 0.51 to 0.33 nats at horizon 1, so every gap reported here is a lower bound on its unclipped value. The composed arm has no coverage guarantee, which is the thing being priced, and it is not proposed as a conformal method. The floor proposition covers invertible signed scores only; folded and quantile scores need their own treatment.

A Proof of Proposition 1

Fix the fold, so (A, κ) is deterministic. Given $X = x$, change of variables gives the true density $p(y|x) = p_W(A(x, y)|x) A'(x, y)$, and the conformal density is $\hat{f}_x(y) = \hat{g}_{\kappa(x)}(A(x, y)) A'(x, y)$ with $\hat{g}_{\kappa}$ the density of the absolutely continuous estimate $\hat{G}_{\kappa}$. The Jacobian cancels in the log ratio, so

$$\text{KL}(P(\cdot|x) \parallel \hat{F}_x) = \text{KL}(P_W(\cdot|x) \parallel \hat{G}_{\kappa(x)}) = \text{KL}(P_W(\cdot|x) \parallel G_{\kappa(x)}) + \mathbb{E}_{W \sim P_W(\cdot|x)} \log \frac{g_{\kappa(x)}(W)}{\hat{g}_{\kappa(x)}(W)}.$$

Averaging over X : the first term is $\mathbb{E}_X \text{KL}(P_W(\cdot | X) \| P_W(\cdot | \kappa(X))) = \text{I}(W; X | \kappa)$ by definition, and the second averages to $\mathbb{E}_\kappa \text{KL}(G_\kappa \| \hat{G}_\kappa) \geq 0$. Every calibration operation is a function of the pooled scores alone, hence acts only on $\hat{G}_\kappa$ and the second term. $\square$

B Reproduction

Every constant regenerates from a tidy store keyed (`task`, `source`, `dataset`, `uid`, `model`, `horizon`, `repr`, `metric`, `value`); the script `constants.py` beside this paper’s source performs the regeneration and fails if the manuscript disagrees with the store. Breadth tier rows are tagged `forecast:CONF:ctx256:.../m6+hash`, deep tier `CONF:ctx1024`, controls `CONFNB/CONFNULL`; the hash covers the harness sources, the tracked diff, and the compiled core. The survey classifier, frame, and per-paper verdicts live beside the grammar paper. Study declarations and the fill protocol are in the `skaters` repository (`surrogate/bench/studies.py`, `skills/run-a-study.md`).

References

- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference* 10(2).
- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *Annals of Statistics* 51(2), 816–845.
- Bogner, K., Pappenberger, F., and Cloke, H. L. (2012). The normal quantile transformation and its application in flood forecasting. *HESS* 16, 1085–1094.
- Cai, T. T. and Low, M. G. (2004). An adaptation theory for nonparametric confidence intervals. *Annals of Statistics* 32(5), 1805–1840.
- Chen, X. and Fan, Y. (2006). Estimation of copula-based semiparametric time series models. *Journal of Econometrics* 130(2), 307–335.
- Chernozhukov, V., Wüthrich, K., and Zhu, Y. (2021). Distributional conformal prediction. *PNAS* 118(48), e2107794118.
- Cotton, P. (2026). Transforms all the way down: automatic online distributional forecasting by conjugation. Working paper, <https://skaters.microprediction.org>.
- Gibbs, I. and Candès, E. (2021). Adaptive conformal inference under distribution shift. *NeurIPS* 34.
- Guan, L. (2023). Localized conformal prediction: a generalized inference framework. *JRSS-B* 85(2).
- Hore, R. and Barber, R. F. (2023). Conformal prediction with local weights: randomization enables local guarantees. arXiv:2310.07850.
- Lei, J. and Wasserman, L. (2014). Distribution-free prediction bands for non-parametric regression. *JRSS-B* 76(1), 71–96.

- Li, K.-C. (1989). Honest confidence regions for nonparametric regression. *Annals of Statistics* 17(3), 1001–1008.
- Low, M. G. (1997). On nonparametric confidence intervals. *Annals of Statistics* 25(6), 2547–2554.
- Rügamer, D., et al. (2023). Probabilistic autoregressive transformation models. *Statistics and Computing*.
- Tibshirani, R. J., Foygel Barber, R., Candès, E. J., and Ramdas, A. (2019). Conformal prediction under covariate shift. *NeurIPS* 32.
- Xu, C. and Xie, Y. (2023). Sequential predictive conformal inference for time series. *ICML*.